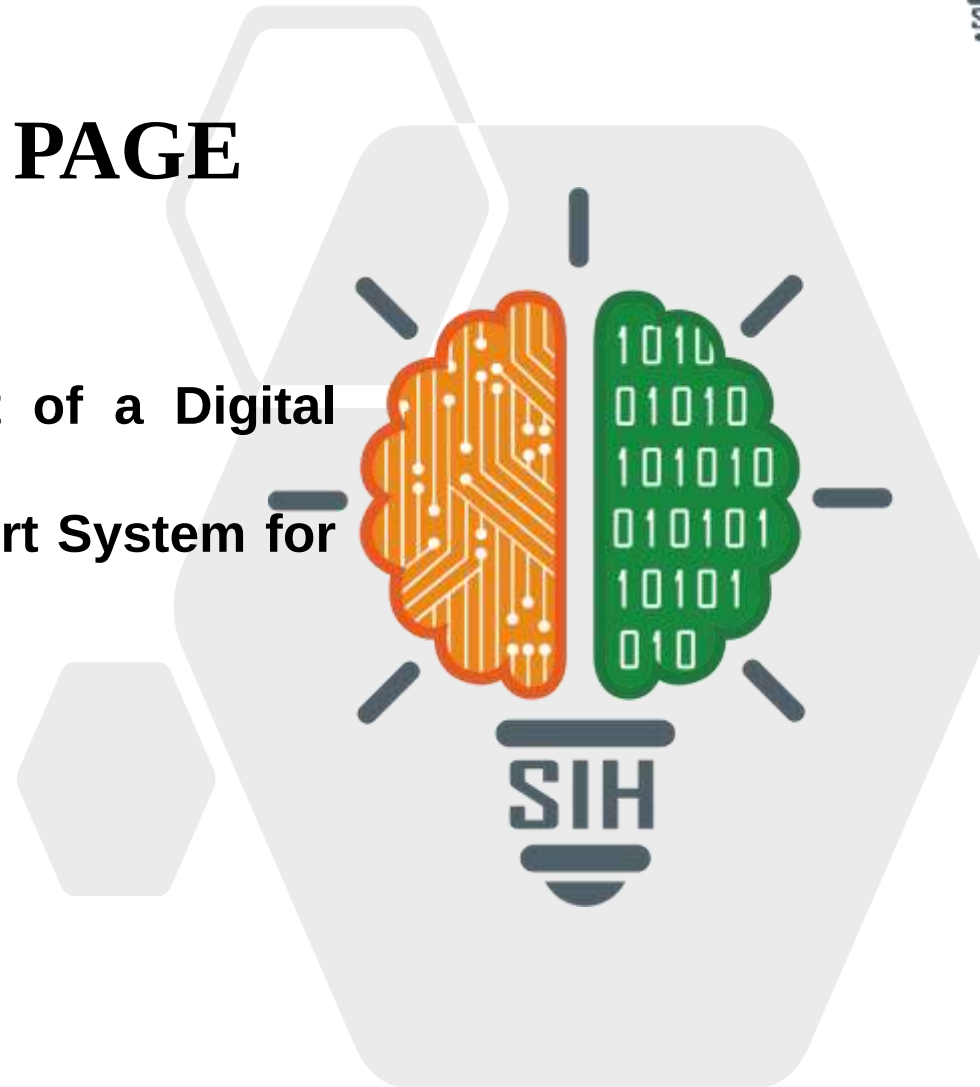


SMART INDIA HACKATHON 2025



TITLE PAGE

- Problem Statement ID – SIH25092
- Problem Statement Title- Development of a Digital Mental Health and Psychological Support System for Students in Higher Education
- Theme- MedTech/BioTech/HealthTech
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)- MIND X





PROBLEMS STATEMENT :

- ❖ **Stigma-Free Access** – Private, judgment-free entry to mental health support.
- ❖ **Early Detection** – AI chatbot + bio-sensors spot distress signs early.
- ❖ **Data-Driven** – EEG provide objective stress insights & coping mechanisms.

PROPOSED SOLUTION :

- ❖ Integrated system combining **psychological support tools** with **psychological stress management**.
- ❖ Provides a **private , accessible, and non-judgemental platform**.
- ❖ Acts as a **first point of contact** for students struggling with mental health.

INNOVATION AND UNIQUENESS:

- ❖ Integration of a **multilingual psychological support platform with cost-effective EEG biofeedback sensors**.
- ❖ Delivers a **personalized, proactive, and data-driven support system**.
- ❖ Strategically **designed** for the **Indian collegiate ecosystem**.
- ❖ Daily **monitoring** and **evaluation**.

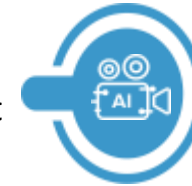
Centralized, Anonymized, Data-Driven Policy Making: Admin Dashboard with Aggregated, Anonymized Analytics

AI-Driven Personalization & Early Intervention: AI-Guided First-Aid Chatbot with Contextual Intelligence

Hyper-Local & Culturally Sensitive Approach: Regional Language Support & Institution-Specific Customization

Holistic, Multi-Layered Support System: Integrated Ecosystem (Chatbot + Booking + Resources + Peer Support + Sensors)

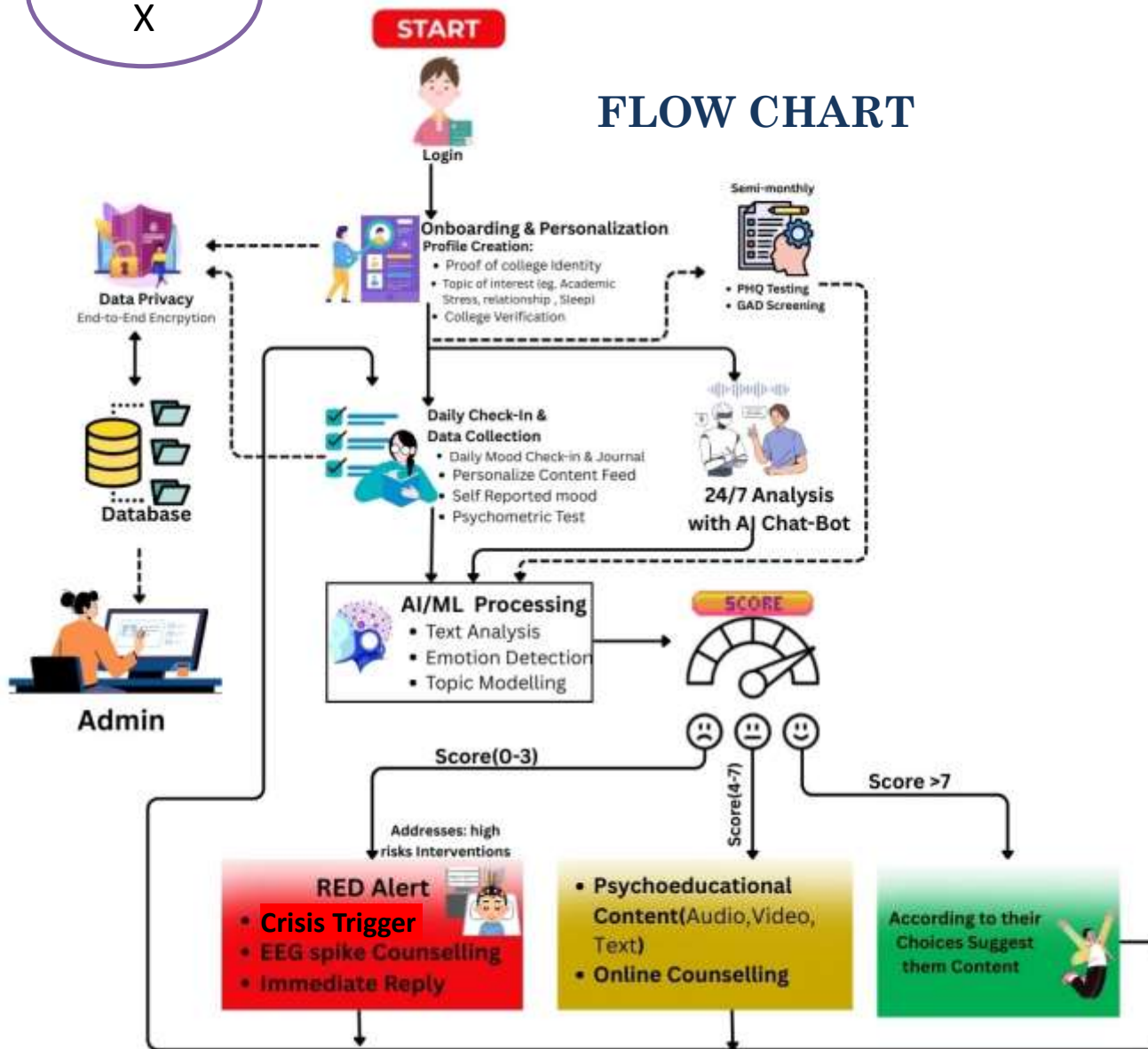
Hardware-Software Integration for Proactive Wellness (The BIG Differentiator): Bio-Physiological Feedback System (EEG Integration)



KEY FEATURES



FLOW CHART



TECH STACK



Technologies to be used:

AI/ML Algorithms :

- Speech-to-text
- Voice Emotion Recognition
- NLP :
 - Sentiment Analysis
 - Emotion Detection
 - Mental health condition detection
 - Topic Modeling

Mobile Applications :

Provides user -friendly interface for real-time feedback

EEG Headband :

Measures brainwave activity for focus and relaxation

ESP32 Microcontroller :

Processes sensor data for noise removal and transmission

FEASIBILITY AND VIABILITY

CHALLENGES	SOLUTION
Privacy	Use strong security, encryption, and clear data policies
Stigma	Run awareness campaigns to normalize mental health help
Accessibility	Add multiple languages, simple design, and disability-friendly features
Low Engagement	Make the app interactive, personalized, and easy to use
Need for Human Support	Connect system with real counsellors for backup
Accuracy	Use reliable tools and regular expert reviews
Technical Issues	Ensure good servers, offline features, and device compatibility
Ethics	Be transparent, give students control over data sharing
Funding	Get support from universities, government, or NGOs
Crisis Handling	Add 24/7 helplines and emergency contact features

FEASIBILITY:

The project is feasible as it uses low-cost sensors and open-source technology, making it affordable and scalable.

- ❑ It is easy to implement, requires minimal training, and has low maintenance needs.
- ❑ Being student-friendly, anonymous, and aligned with national policies, it is socially acceptable and sustainable.

VIABILITY:

- ❑ Institution subscription (per student/year).
- ❑ Freemium model for students (basic free, advanced insights premium).
- ❑ Can expand beyond higher education → to **schools, workplaces, corporates**.
- ❑ Continuous improvement → AI models improve as more data is collected.



POTENTIAL IMPACT ON STUDENTS

**Education**

- Provides a scalable solution to support thousands of students.
- Aims to improve academic performance and reduce dropout rates linked to mental health issues.
- Provides data-driven academic policies for colleges.

**Healthcare**

- Acts as a preventive mental health system—reducing hospital/therapy load.
- Supports early screening for anxiety, depression, and burnout.
- Helps counselors manage larger student population.

**Social**

- Normalizes conversations around mental health on campus.
- Fosters a supportive community through the moderated peer-support platform.
- Breaks mental health stigma among youth.

EMPOWERMENT

“ Gives students tools to understand their mental and physiological states and actively manage their well-being. ”

24/7 ACCESSIBILITY

“ Provides immediate, round-the-clock first-aid support, which is crucial for students facing sudden distress. ”

REDUCED STIGMA

“ Offers a confidential way to seek help, encouraging early intervention before problems escalate. ”

➤ **Chan, A. H. Y., & Honey, M. (2021).**

User perceptions of mobile digital apps for mental health: Acceptability and usability-An integrative review. **Journal of Psychiatric and Mental Health Nursing**,29(1), 147-168.

➤ **Patel, S., Akhtar, A., Malins, S., Wright, N., Rowley, E., Young, E., Sampson, S., Morriss, R. (2020).**

The acceptability and usability of digital health interventions for adults with depression, anxiety and disorders: Qualitative systematic review and meta-synthesis.
Journal of Medical Internet Research,22(7),e16228.

➤ **Lattie, E. G., Adkins, E. C., Winkvist, N., Stiles-Shields, C., Wafford, Q. E., & Graham, A. K. (2019).**

Digital mental health interventions for depression, anxiety, and enhancement of psychological well-being among college students: Systematic review. **Journals of Medical Internet Research**,21(7),e12869.

➤ **An-Di Liang et al. Sheng Li Xue Bao. (2022).**

Study on the EEG functional brain connectivity characteristics of college students with depression.

SET OF QUESTIONS IN PSYCHOMETRIC TEST

1. Daily Life & Emotional Reactions

- When your day doesn't go as planned, how do you usually respond?
- When plans suddenly change, how do you adapt and what feelings arise?
- During stressful times, where do you notice anxiety the most?
- When I woke up this morning, what was the very first feeling I noticed in my mind or body?

2. Stressful Situations

- you feel emotionally drained but still have responsibilities — what do you do?
- You're facing multiple deadlines with limited time — what's your plan?
- You get negative feedback from your boss or teacher — what do you do?

3. Problem-Solving & Social Situations

- Something unexpected disrupts your plans — how do you adjust?
- When nothing seems to work, what helps you keep going?
- Everyone around you disagrees with your opinion — how do you respond?
- You're the only one standing up for something you believe in — how do you feel?
- You feel left out of a group decision — what do you do?

❖ Further alternatives can be provided, contingent upon the specific scenario.